

Class 11: Protein Structure Prediction with AlphaFold

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Background

We saw last day that the main repository for biomolecular structure (the PDB database) only has ~250,000 entries.

UniProtKB (the main protein sequence database) has over 200 million entries!

AlphaFold

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021)

Without the aid of such approaches it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

The EBI AlphaFold database

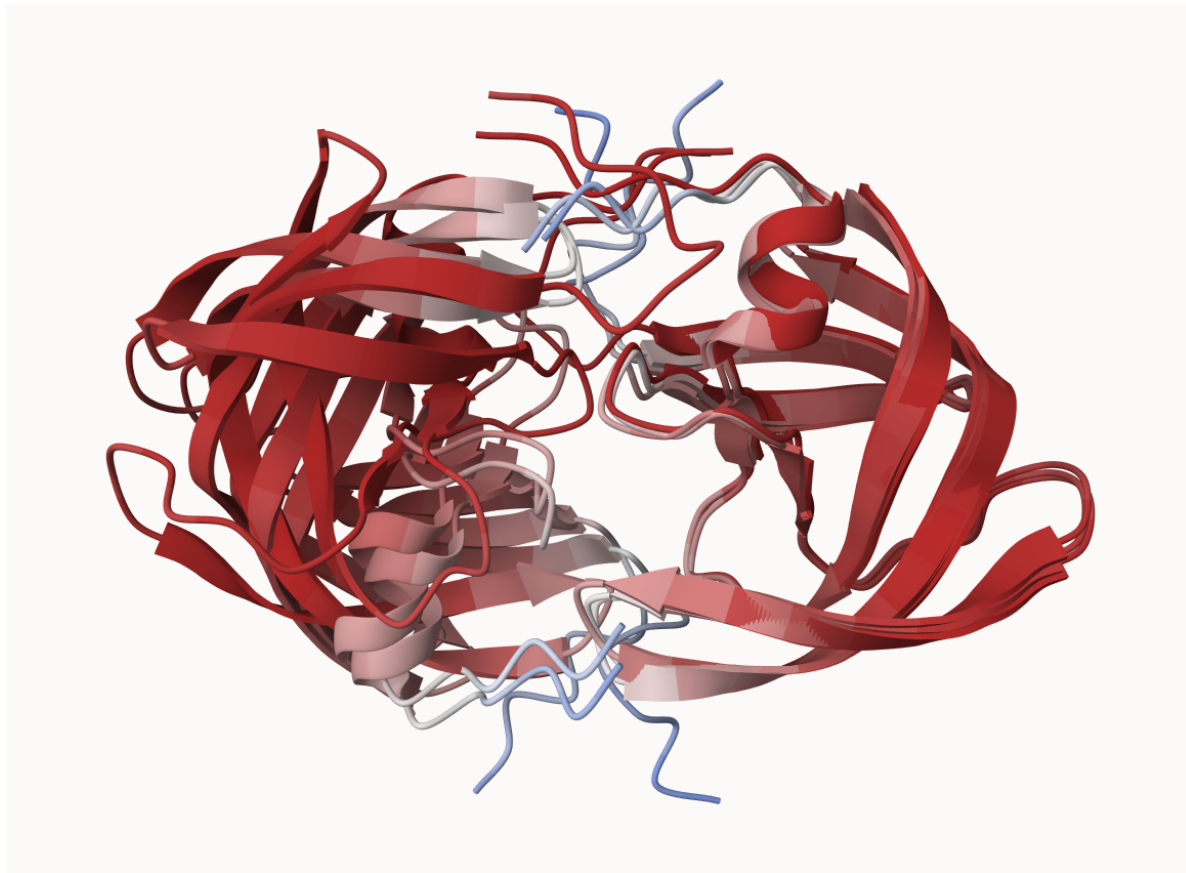
The EBI AlphaFold database contains lots of computed structure models. It is increasingly likely that the structure you are interested in is already in this database.

There are 3 major outputs from AlphaFold

1. A model of structure in **PDB** format
2. a **pLDDT score**: that tells us how confident the model is for a given residue in your protein (High values are good, above 70)
3. a **PAE score** that tells us about protein packing quality

If you can't find a matching entry for the sequence you are interested in AFDB you can run AlphaFold yourself...

Running AlphaFold



We will use Colab

Interpreting Results

Custom analysis of resulting models

We can read all the AlphaFold results into R and do more quantitative analysis than just viewing the structures in Mol-star

Read all the PDB models:

```
library(bio3d)
p <- read.pdb ("hivpr_23119 2")
```

```
list.files("hivpr_23119 2/", pattern = ".pdb", full.names = T)
```

```
[1] "hivpr_23119 2//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_23119 2//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_23119 2//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_23119 2//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_23119 2//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

Custom analysis of resulting models

```
# Change this for YOUR results dir name
results_dir <- "hivpr_23119 2/"
```

```
# File names for all PDB models
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)
```

```
# Print our PDB file names
basename(pdb_files)
```

```
[1] "hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
#library(bio3dview)
#view.pdbs(pdbs)
```

How similar or different are my models?

```
library(bio3d)

# Read all data from Models
# and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
hivpr_23119 2//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119 2//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119 2//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119 2//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119 2//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....
```

Extracting sequences

```
pdb/seq: 1   name: hivpr_23119 2//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
pdb/seq: 2   name: hivpr_23119 2//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
pdb/seq: 3   name: hivpr_23119 2//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
pdb/seq: 4   name: hivpr_23119 2//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
pdb/seq: 5   name: hivpr_23119 2//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
```

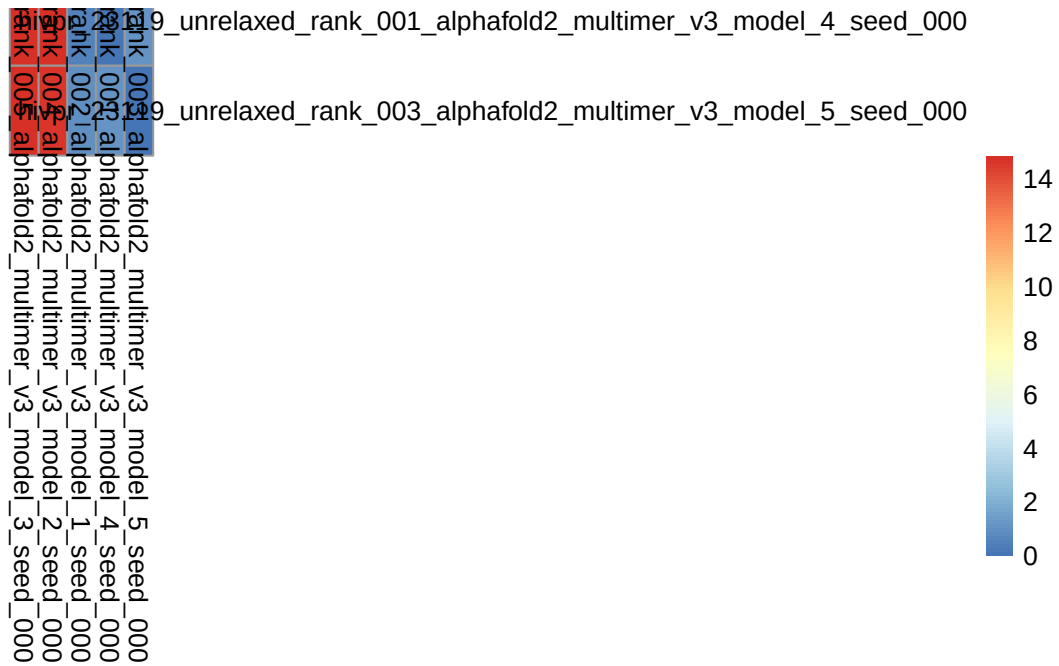
```
rd<- rmsd(pdbs, fit=T)
```

Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions

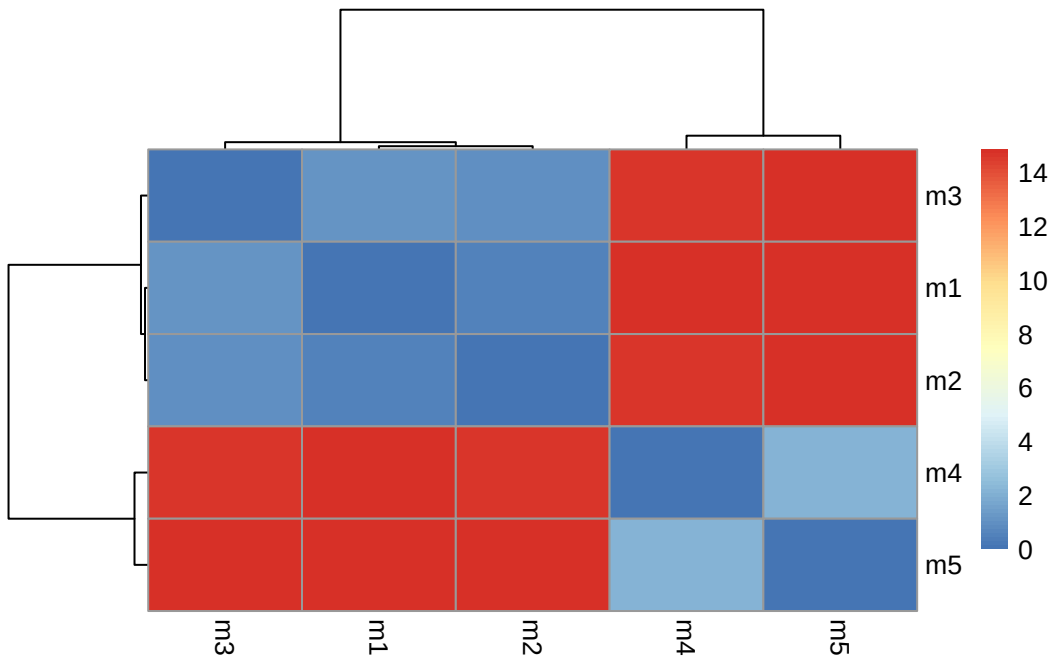
```
range(rd)
```

```
[1] 0.000 14.851
```

```
library(pheatmap)
pheatmap(rd)
```



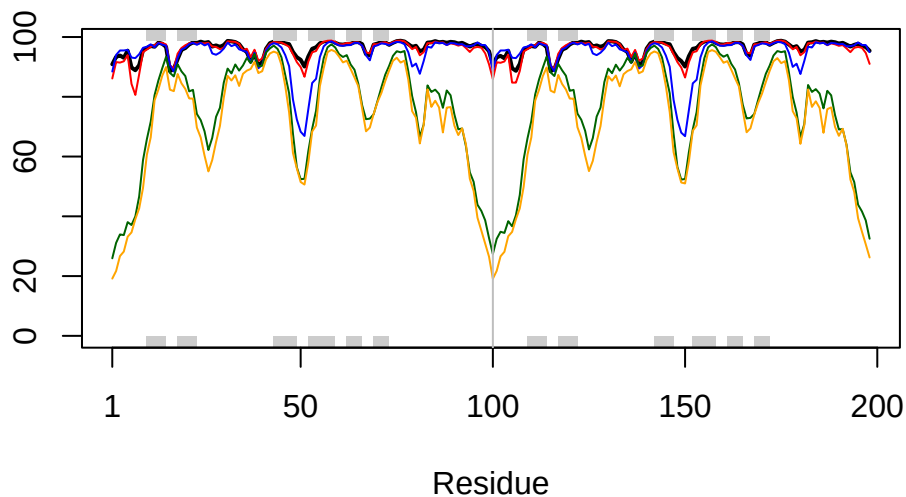
```
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



```
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
plotb3(pdb$b[1,], typ="l", lwd=2, sse=pdb)
points(pdb$b[2,], typ="l", col="red")
points(pdb$b[3,], typ="l", col="blue")
points(pdb$b[4,], typ="l", col="darkgreen")
points(pdb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



```
core <- core.find(pdb)
```

```
core size 197 of 198 vol = 558.878
core size 196 of 198 vol = 515.506
core size 195 of 198 vol = 486.798
core size 194 of 198 vol = 477.921
core size 193 of 198 vol = 466.233
core size 192 of 198 vol = 458.023
core size 191 of 198 vol = 448.272
```

core size 190 of 198	vol = 440.017
core size 189 of 198	vol = 433.278
core size 188 of 198	vol = 426.693
core size 187 of 198	vol = 420.962
core size 186 of 198	vol = 415.107
core size 185 of 198	vol = 410.888
core size 184 of 198	vol = 408.533
core size 183 of 198	vol = 407.448
core size 182 of 198	vol = 406.487
core size 181 of 198	vol = 405.084
core size 180 of 198	vol = 390.831
core size 179 of 198	vol = 378.472
core size 178 of 198	vol = 374.296
core size 177 of 198	vol = 368.302
core size 176 of 198	vol = 361.133
core size 175 of 198	vol = 354.435
core size 174 of 198	vol = 347.289
core size 173 of 198	vol = 340.332
core size 172 of 198	vol = 332.772
core size 171 of 198	vol = 325.968
core size 170 of 198	vol = 318.125
core size 169 of 198	vol = 310.342
core size 168 of 198	vol = 304.618
core size 167 of 198	vol = 296.047
core size 166 of 198	vol = 286.968
core size 165 of 198	vol = 278.853
core size 164 of 198	vol = 270.427
core size 163 of 198	vol = 262.291
core size 162 of 198	vol = 254.155
core size 161 of 198	vol = 246.894
core size 160 of 198	vol = 240.783
core size 159 of 198	vol = 231.122
core size 158 of 198	vol = 225.244
core size 157 of 198	vol = 218.539
core size 156 of 198	vol = 212.033
core size 155 of 198	vol = 203.964
core size 154 of 198	vol = 195.317
core size 153 of 198	vol = 188.3
core size 152 of 198	vol = 184.235
core size 151 of 198	vol = 175.942
core size 150 of 198	vol = 170.395
core size 149 of 198	vol = 165.371
core size 148 of 198	vol = 159.598

core size 147 of 198	vol = 154.385
core size 146 of 198	vol = 150.936
core size 145 of 198	vol = 147.281
core size 144 of 198	vol = 141.906
core size 143 of 198	vol = 137.092
core size 142 of 198	vol = 131.182
core size 141 of 198	vol = 126.228
core size 140 of 198	vol = 124.052
core size 139 of 198	vol = 121.83
core size 138 of 198	vol = 117.678
core size 137 of 198	vol = 112.571
core size 136 of 198	vol = 109.476
core size 135 of 198	vol = 106.032
core size 134 of 198	vol = 103.025
core size 133 of 198	vol = 98.215
core size 132 of 198	vol = 94.252
core size 131 of 198	vol = 90.004
core size 130 of 198	vol = 86.323
core size 129 of 198	vol = 81.501
core size 128 of 198	vol = 78.646
core size 127 of 198	vol = 74.276
core size 126 of 198	vol = 71.847
core size 125 of 198	vol = 68.247
core size 124 of 198	vol = 63.98
core size 123 of 198	vol = 60.725
core size 122 of 198	vol = 57.254
core size 121 of 198	vol = 55.565
core size 120 of 198	vol = 52.591
core size 119 of 198	vol = 49.711
core size 118 of 198	vol = 47.588
core size 117 of 198	vol = 45.962
core size 116 of 198	vol = 42.882
core size 115 of 198	vol = 40.953
core size 114 of 198	vol = 38.767
core size 113 of 198	vol = 35.926
core size 112 of 198	vol = 33.704
core size 111 of 198	vol = 31.761
core size 110 of 198	vol = 29.581
core size 109 of 198	vol = 28.135
core size 108 of 198	vol = 26.192
core size 107 of 198	vol = 24.771
core size 106 of 198	vol = 23.278
core size 105 of 198	vol = 21.757


```

core size 104 of 198  vol = 20.456
core size 103 of 198  vol = 18.869
core size 102 of 198  vol = 17.997
core size 101 of 198  vol = 16.068
core size 100 of 198  vol = 14.328
core size 99 of 198   vol = 13.166
core size 98 of 198   vol = 12.138
core size 97 of 198   vol = 10.992
core size 96 of 198   vol = 9.745
core size 95 of 198   vol = 9.229
core size 94 of 198   vol = 7.852
core size 93 of 198   vol = 6.687
core size 92 of 198   vol = 5.073
core size 91 of 198   vol = 4.181
core size 90 of 198   vol = 3.547
core size 89 of 198   vol = 2.205
core size 88 of 198   vol = 1.731
core size 87 of 198   vol = 1.339
core size 86 of 198   vol = 1.051
core size 85 of 198   vol = 0.777
core size 84 of 198   vol = 0.456
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol=0.5)
```

```

# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1      8 49      42
2     53 95      43

```

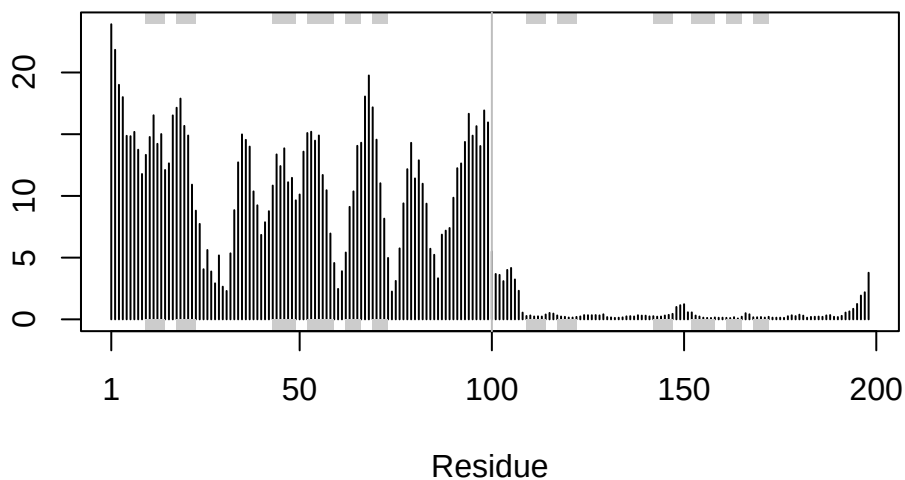
```
xyz <- pdbfit(pdb, core.inds, outpath="corefit_structures")
```

```
rf <- rmsf(xyz)
```

```

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")

```



```
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)

pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)

$names
[1] "plddt"    "max_pae" "pae"      "ptm"      "iptm"

# Per-residue pLDDT scores
# same as B-factor of PDB..
head(pae1$plddt)

[1] 90.88 93.38 93.81 93.06 95.44 89.75
```

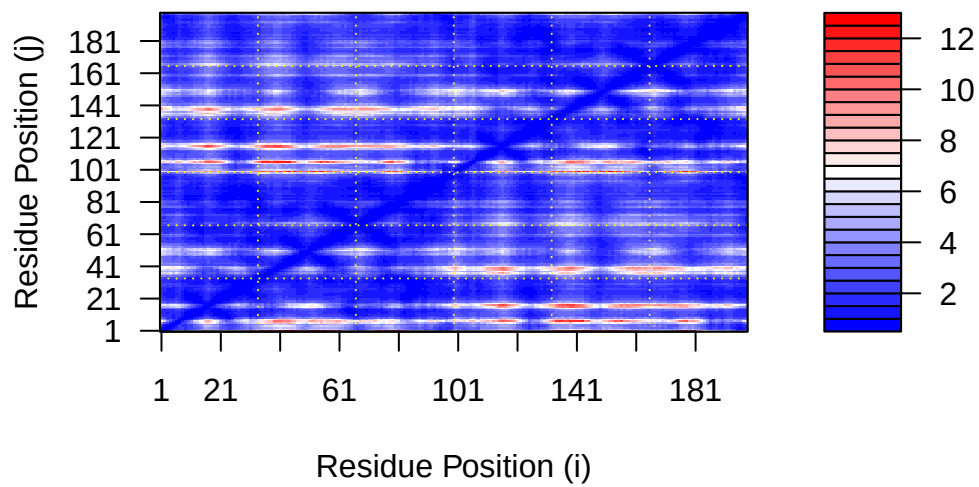
```
pae1$max_pae
```

```
[1] 12.89062
```

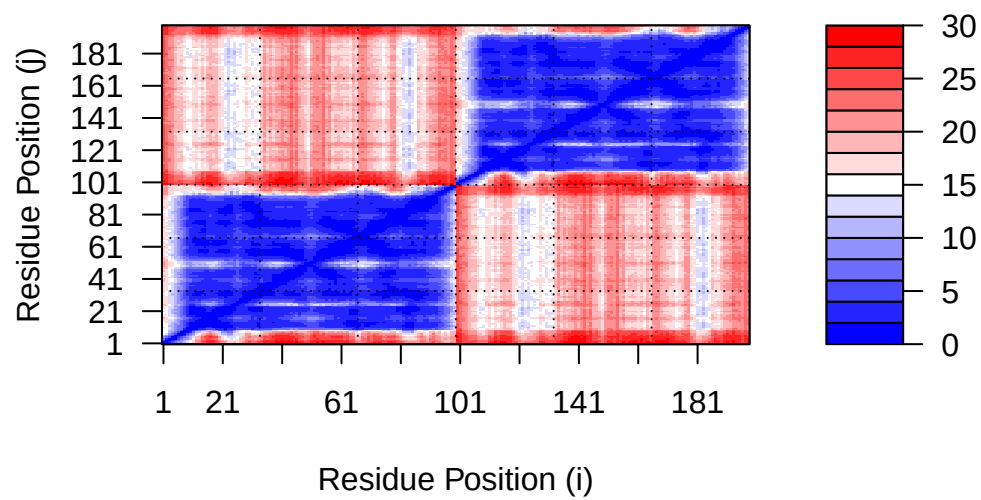
```
pae5$max_pae
```

```
[1] 29.54688
```

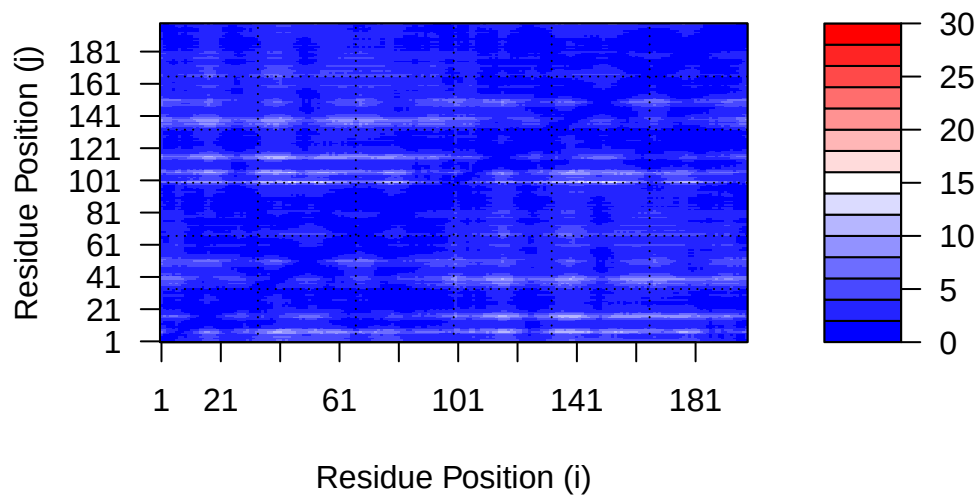
```
plot.dmat(pae1$pae,  
          xlab="Residue Position (i)",  
          ylab="Residue Position (j)")
```



```
plot.dmat(pae5$pae,  
          xlab="Residue Position (i)",  
          ylab="Residue Position (j)",  
          grid.col = "black",  
          zlim=c(0,30))
```



```
plot.dmat(pae1$pae,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                       full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119 2//hivpr_23119.a3m"
```

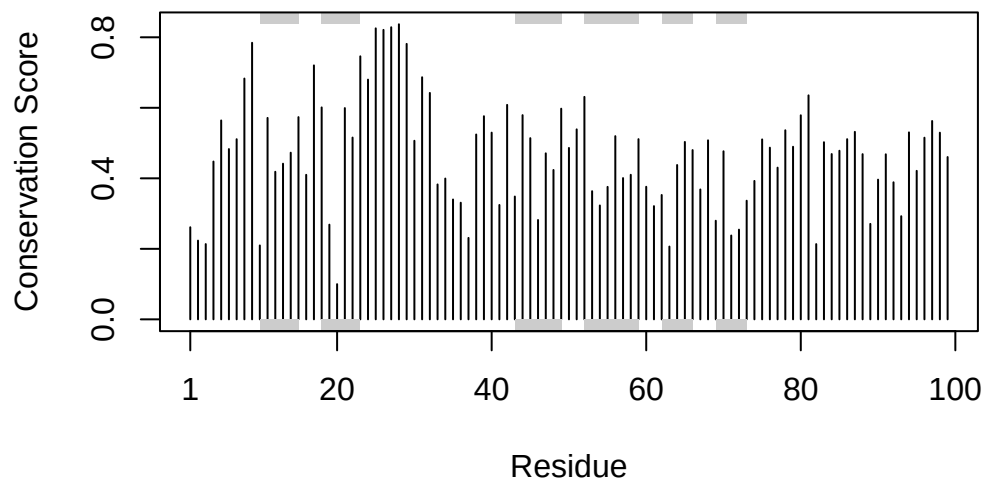
```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397 132
```

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```